RAMIRO YAIR ROMÁN TREJO

Junior Backend Developer (Python / Django)

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WORK EXPERIENCE

Software Developer Intern — Compañía Cervecera de Zacatecas (Grupo Modelo / AB InBev) Aug 2026 – Oct 2026

- Digitized tool auditing and Good Operating Practices (GOPs) inspections in the Packaging Department, replacing manual paper logs with a centralized web application.
- Built the backend and relational database schema using **Python (Django)** and **PostgreSQL**, automating issue tracking and corrective action plan generation on the plant floor.
- Containerized the application with **Docker Compose** for consistent deployment and connected operational records with maintenance KPI reports (**MTTR and LEF**) in Power BI.

Quality Assurance (QA) Inspector — Nagakura Engineering Works Apr 2021 – Jul 2022

- Executed dimensional and material inspection protocols across automotive manufacturing lines under strict quality tolerances.
- Documented non-conformances and performed **Root Cause Analysis (RCA)** with the engineering team to implement corrective actions and reduce production line defects.

SOFTWARE DEVELOPMENT PROJECTS

Academic Curriculum Tracking Platform (UAZ – Lic. en Lenguas Extranjeras) | Python, Django, MariaDB, Jenkins, Docker Nov 2025 – May 2026

- Developed the backend for an institutional web platform to manage course syllabi and faculty tracking, replacing manual spreadsheet workflows.
- Configured automated testing and deployment to a Linux server on **AWS EC2** using **Jenkins** and **Docker**, integrating **Behave (BDD)** and **Selenium** tests to verify core user flows prior to release.

Crop Pathology Detection System (Doctor Chile) | Python, PyTorch, YOLOv8, Flutter, SQLite github.com/YairRT45/doctor_chile_app

- Trained a computer vision model (**YOLOv8**) to classify 5 foliar pests and diseases in chili crops from field photographs.
- Built a mobile application in **Flutter** with local **SQLite** storage, enabling farmers to run diagnostics and track crop history offline in rural areas.

University Enrollment & Scheduling System (UPN-SASCE) | Python, Django, PostgreSQL, Docker Institutional Project

- Implemented student enrollment and class scheduling modules for **Universidad Pedagógica Nacional (UPN)**, validating course prerequisites and capacity rules through automated **Behave** tests.

Data Structures & Graph Algorithms in Java | Java github.com/YairRT45

- Implemented core data structures from scratch (Binary Search Trees, Heaps, Directed Graphs) and **Dijkstra's Algorithm** for shortest-path routing, along with a stack-based syntax parser.

TECHNICAL SKILLS

Languages & Backend: Python, Django, Django REST Framework (DRF), FastAPI, SQL, Java.

Databases & Infrastructure: PostgreSQL, MariaDB / MySQL, SQLite, Docker, Docker Compose, Linux (Ubuntu / Debian, Bash), Nginx.

Tools, CI/CD & Testing: Git, GitHub Actions, Jenkins, BDD Testing (Behave), Selenium, pytest, Basic Monitoring (Prometheus / Grafana).

EDUCATION & LANGUAGES

B.S. in Software Engineering — Universidad Autónoma de Zacatecas (UAZ) 2022 – Present (Final Semester)

Languages: Spanish (Native) • English (B1 – Intermediate: technical documentation reading and written communication).